

Class Imbalance in Histopathology

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1 Motivation

Histopathology datasets are long-tailed almost by construction. Common cancers and benign tissue dominate archives, while rare subtypes and atypical lesions, often the diagnostically decisive ones, contribute few patients and patches. BRACS, for example, was released with explicitly unbalanced atypical classes (Brancati et al., 2022). In CNNs, imbalance biases predictions toward frequent classes and lowers minority recall (Buda et al., 2018); in medical imaging it also distorts calibration, and conclusions depend on whether evaluation uses balanced metrics (Mosquera et al., 2024). Long-tailed medical benchmarks show that no single remedy works across datasets (Ju et al., 2024), and histopathology-specific models keep being proposed (Cong et al., 2022). Frozen foundation-model features with a linear probe are now the standard baseline (Chen et al., 2024; Zimmermann et al., 2024), yet how much imbalance still costs such a probe, and why, has not been measured in isolation. This series answers both questions before mitigation methods are tested.

2 Setup

A multinomial logistic regression is trained on frozen Virchow2 patch features (Zimmermann et al., 2024) on two tasks: TCGA-UT, 30 cancer types (Komura et al., 2022), and BRACS, 7 breast-lesion subtypes from normal tissue to invasive carcinoma (Brancati et al., 2022). Every condition uses the same patients per class (20 on TCGA-UT, 10 on BRACS) and the same total patch budget; only the share of that budget each class receives changes. An exponential profile sets the ratio ρ between the largest and smallest class to 1–100 (Figure 1), class-to-rank assignment is randomized over 30 draws (3 splits $\times$ 10), and a native condition keeps each dataset’s own class shares. The outcome is the *damage* $\Delta(\rho) = \text{BA}(1) - \text{BA}(\rho)$, the loss in test patient-macro balanced accuracy against the balanced condition, with paired bootstrap 95% intervals; damage counts as material from 1 percentage point (pp).

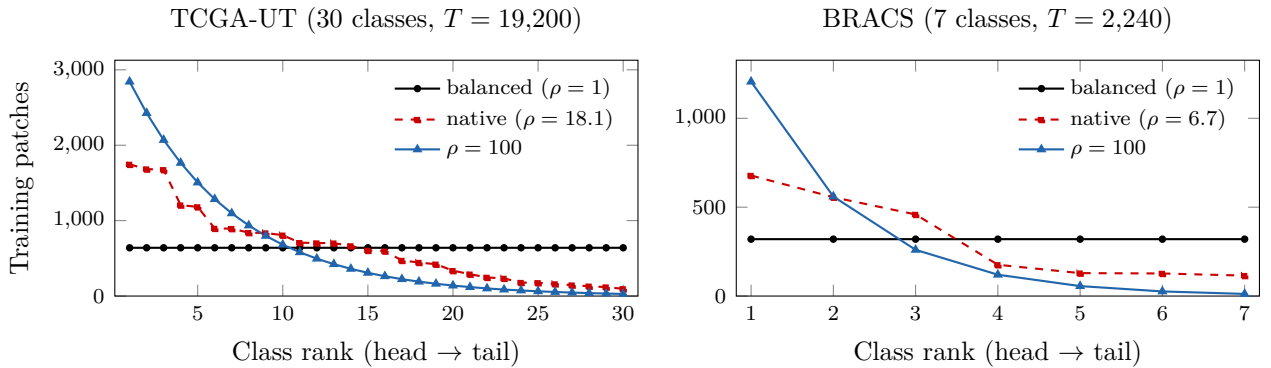


Figure 1: Training class distributions, classes sorted from head to tail: the balanced condition, the native class shares of one split, and the most imbalanced ratio condition. The tested ratios are $\rho \in \{1, 2, 5, 10, 20, 50, 100\}$; the profiles for $\rho = 2$ to 50 lie between the balanced and the $\rho = 100$ profile. Which class sits at which rank is randomized per draw.

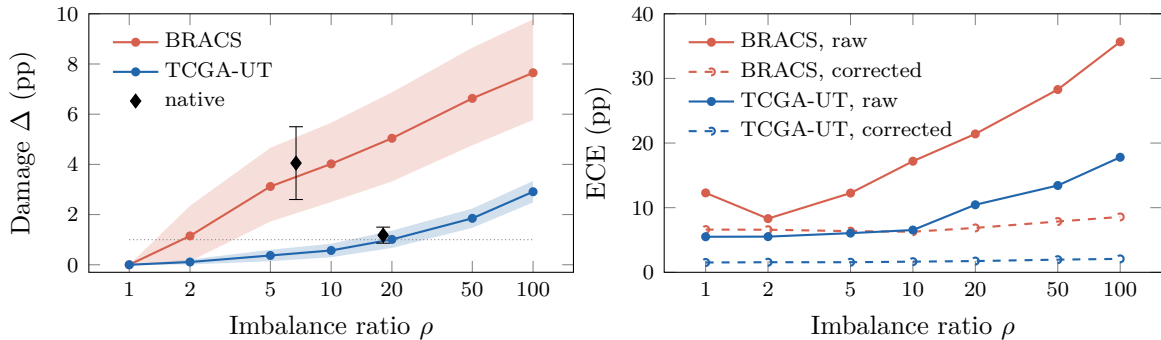


Figure 2: Left: balanced-accuracy damage versus ρ with 95% bands; diamonds mark native class shares, dotted line the 1-pp threshold. Right: expected calibration error (ECE) of the raw predicted probabilities (solid) and after a single global confidence correction fitted on validation data (dashed).

3 Effects of class imbalance

The balanced conditions reach 72.1% balanced accuracy on TCGA-UT and 39.6% on BRACS. On TCGA-UT, published ROI-level linear evaluations of 32 classes score 0.60–0.68 for current encoders (Virchow2 0.62, UNI2-h 0.68; Mahmood Lab, 2025); the higher value here reflects 30 classes and aggregation over each patient’s patches. For seven-class BRACS subtyping, the CNN baseline of Brancati et al. (2022) reaches 55.7% ROI-level balanced accuracy, calculated as the mean of the seven class recalls in their Table 9. Vaidya et al. (2025) report slide-level balanced accuracy of 34.8% with mean-pooled Virchow features, 41.9% with PRISM, and 48.1% with THREADS, all using linear probes and five patient-disjoint splits of the 189-patient cohort. The 39.6% here lies within the reported slide-level range, but this study computes patient-macro recall from patch predictions. Different training budgets and evaluation units make these results context for task difficulty, not matched performance benchmarks.

Imbalance damage differs strongly between the two tasks (Figure 2, left). On TCGA-UT it is small and accelerating: 0.57 pp at $\rho = 10$, the 1-pp threshold only at $\rho = 20$, and 2.91 pp [2.48, 3.34] at $\rho = 100$. On BRACS it starts immediately and grows steadily by about 1.15 pp per doubling of ρ : 1.15 pp already at $\rho = 2$, 4.02 pp at $\rho = 10$, and 7.65 pp [5.77, 9.78] at $\rho = 100$, a fifth of the balanced level. Both native allocations lie on their curves (1.18 and 4.05 pp). The aggregate hides a large shift of recall from tail to head: at $\rho = 100$, the TCGA-UT tail third loses 11.8 pp while the head gains 5.4 pp; on BRACS the two tail classes drop from 38.7% to 7.3% recall. The predicted probabilities suffer far more than accuracy (Figure 2, right). The expected calibration error (ECE), the average gap between the model’s confidence and its accuracy, rises from 5.5 to 17.8 pp on TCGA-UT and from 12.3 to 35.7 pp on BRACS at $\rho = 100$: the model becomes overconfident, but uniformly so. A global confidence correction that divides all logits by one scalar fitted on validation data (temperature scaling; Guo et al., 2017) removes 92–96% of the increase.

4 Class influence and tail support

When a tail class receives fewer training patches, two things change. Its errors count less often in the training loss, so common classes have more influence on the fitted decision boundary. This is the *prior* effect: it concerns how strongly the available examples of each class shape the classifier. The model also sees fewer distinct tail patches, at $\rho = 100$ only one or two per patient, so it has less evidence about how that class varies. This is the *support* effect: giving those few patches more weight cannot supply the missing examples. The conditions in Table 1 separate these effects. P keeps all patches from the balanced condition but gives head classes more weight, isolating the prior effect. S keeps the sparse patches from the imbalanced condition but weights classes equally, isolating the support effect. The interaction $I = \Delta_r - \Delta_p - \Delta_s$ measures any additional damage when both occur together.

The two tasks break down differently. On TCGA-UT the prior carries the damage at moderate ratios (0.73 of 1.01 pp at $\rho = 20$) while thin support alone costs nothing; at $\rho = 100$ the two compound, with

	Many tail patches	Few tail patches
Equal class influence	balanced condition (reference)	S: support only
Head-favoured influence	P: prior only	r: ratio condition (both)

Table 1: Four training conditions separate the effect of unequal class influence (prior) from the effect of fewer tail examples (support) at each ρ .

prior 1.20, support 0.53 and an interaction of 1.18 pp. On BRACS both factors are material and add up without resolved interaction: prior 4.47 pp and support 3.29 pp at $\rho = 100$. Thin support alone thus costs BRACS more than the entire imbalance damage on TCGA-UT.

5 Dependence on tail-class identity

Damage depends on which classes end up in the tail, not only on ρ . On BRACS, rotating every class through every rank shows that easy classes cost most when rare: damage at $\rho = 100$ ranges from 5.73 pp (UDH in the tail) to 9.99 pp (invasive carcinoma), because a class with high balanced recall has the most to lose. On TCGA-UT, a class is especially vulnerable when it receives few patches but a class it is often mistaken for receives many. A score fixed before testing new class orders used this pattern to predict their damage. The order ranked most damaging costs 4.51 pp, the least damaging 1.34 pp, and random orders 2.91 pp (Figure 3). Reversing the most damaging order keeps the same rank gaps between pairs of classes but lowers damage by 1.11 pp [0.47, 1.73]. Damage is therefore greatest when the class prone to being mistaken for another is the rarer one.

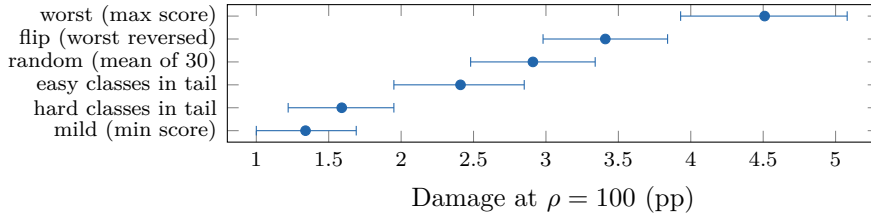


Figure 3: TCGA-UT damage at $\rho = 100$ for different class-to-rank assignments, with 95% intervals.

6 Encoder comparison

Refitting the prior, support and ratio conditions on identical patients with UNI2-h (Chen et al., 2024) instead of Virchow2 leaves the picture intact. On TCGA-UT, UNI2-h starts 1.78 pp higher and loses 0.86 pp less at $\rho = 100$ (97.5% interval -1.33 to -0.40), a resolved but sub-1-pp difference driven by smaller support and interaction damage. On BRACS the difference is unresolved and the tail classes collapse to about 9% recall with both encoders.

7 Implications for mitigation

- Rebalancing the training objective removes the prior damage but cannot restore thin support. On BRACS, 3.3 pp remain at $\rho = 100$; recovering them needs more tail patches or patients, or methods that improve tail-class estimation.
- Most calibration damage is removed by the global confidence correction, so a mitigation method must beat a corrected baseline to add value in probability quality.
- Dataset and class order change the damage by a factor of 2–3 at the same ρ ; evaluations should randomize class order and report both datasets.

A Guide to the supporting experiments

The numbered folders below are under `experiments/03_class_imbalance/`. Each folder’s `report/` directory contains a PDF and its LaTeX source. The reports give the protocols, full tables, and uncertainty estimates behind the results condensed here.

Headline results.

- **Imbalance curves and probability quality:** Reports 25 and 26, in `00_damage/25_damage_tcga_ut/report/` and `00_damage/26_damage_bracs/report/`, give the balanced baselines, damage at every ratio, native allocations, head-to-tail recall, and calibration before and after temperature scaling.
- **Class influence versus tail support:** Reports 27 and 28, in `01_cause/tcga_ut/27_prior_vs_support/report/` and `01_cause/bracs/28_prior_vs_support/report/`, show the paired prior-only and support-only conditions, their interaction, and its uncertainty on each dataset.
- **Which classes become rare:** Report 29, in `01_cause/bracs/29_tail_assignment/report/`, rotates every BRACS class through the tail. Report 32, in `01_cause/tcga_ut/32_worst_assignment_rate/report/`, tests predicted damaging and mild class orders on TCGA-UT, including reversal of the most damaging order.
- **Encoder comparison:** Report 39, in `01_cause/cross/39_encoder_transfer/report/`, compares Virchow2 and UNI2-h on matched patients and patches and examines sensitivity to regularization selection.

Further reading. Reports 25 and 26 also track negative log-likelihood and fitted confidence corrections across the full imbalance grid. Reports 29 and 32 give class-level recall and confusion analyses behind the aggregate class-order effects, useful when the identity of a rare class matters. Report 39 adds per-class encoder comparisons and a fixed-regularization analysis, showing how the probe-selection rule affects the encoder contrast.

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